

1. Code Question 1

Amazon's software team utilizes several algorithms to maintain data integrity, one of which targets the encoding of symmetrical names. Symmetrical names are unique in that they read identically in both directions, similar to palindromes in language parlance.

The chief aim of the algorithm is to rearrange the characters in the original symmetrical name according to these criteria:

- The rearranged name is a reshuffled version of the original symmetrical name.
- The restructured name should be symmetrical as well.
- This restructured name should be lexicographically smallest among all its symmetric permutations.

Given an initial symmetrical name that contains only lowercase English characters, compute the encoded name.

A string s is considered to be lexicographically smaller than the string t of the same length if the first character in s that differs from that in t is smaller. For example, "abcd" is lexicographically smaller than "abdc" but larger than "abad"

Note that the output encoded name could match the original name if it's already the smallest lexicographically.

Example

The original string is *letters = "babab"*.

This can be reversed to give "abbba", which is a symmetric rearrangement of the original symmetrical name and is the smallest possible reverse order.

It satisfies all the requirements so return the string *abbba*.

Function Description

Complete the function *getEncodedName* in the editor below.

getEncodedName has the following parameter:

string letters: the initial symmetrical string name.

Returns

string: the encoded *letters*

Constraints

- $1 \leq |letters| \leq 10^5$
- *letters* consists of lowercase English letters only.
- *letters* is a palindrome.

▼ Input Format For Custom Testing

The first line contains a string, *letters*, the string name.

▼ Sample Case 0

Sample Input For Custom Testing

STDIN		FUNCTION
-----		-----
yxyx	→	letters = "yxyx"

Sample Output

xyyx

Explanation

Rearrange the original letters to generate "xyyx", which is a palindrome and also the smallest possible reverse order.

▼ Sample Case 1

Sample Input For Custom Testing

STDIN		FUNCTION
-----		-----
ded	→	letters = "ded"

Sample Output

ded

Explanation

The *letters* string already is the smallest possible reverse order in lexicographic terms.

```
19 def getEncodedName(letters):
20     # Write your code here
21     freq = Counter(letters)
22     first_half = []
23     middle_char = ""
24
25     for char in sorted(freq.keys()):
26         count = freq[char]
27         if count % 2 == 1:
28             if middle_char:
29                 return ""
30             middle_char = char
31         first_half.append(char * (count // 2))
32
33     first_half_str = "".join(first_half)
34     return first_half_str + middle_char + first_half_str[::-1]
35
36 if __name__ == '__main__':
37     fptr = open(os.environ['OUTPUT_PATH'], 'w')
38
39     letters = input()
40
41     result = getEncodedName(letters)
42
43     fptr.write(result + '\n')
44
45     fptr.close()
```